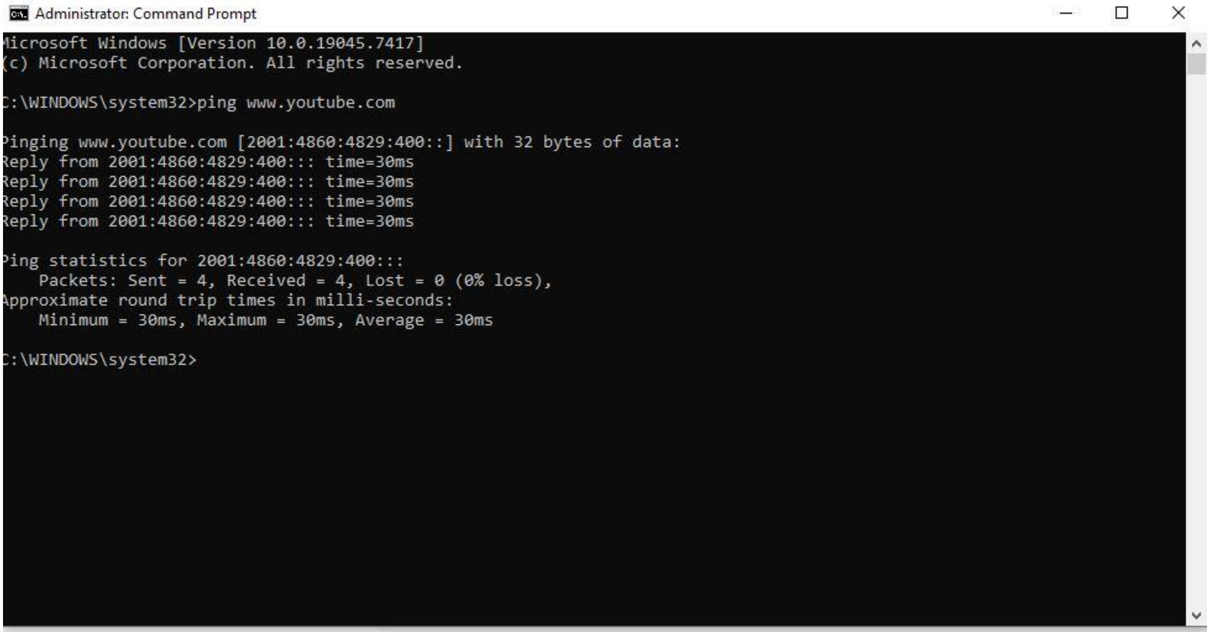
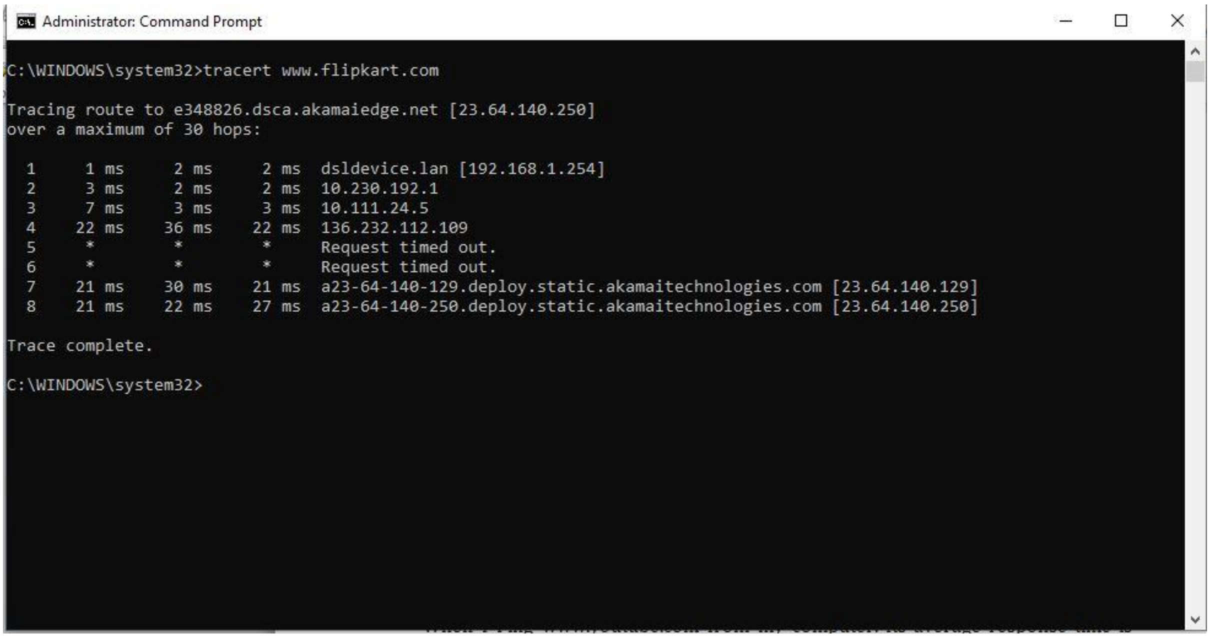
1. **Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.**



- When I Ping www.youtube.com from my computer. Its average response time is 30ms.

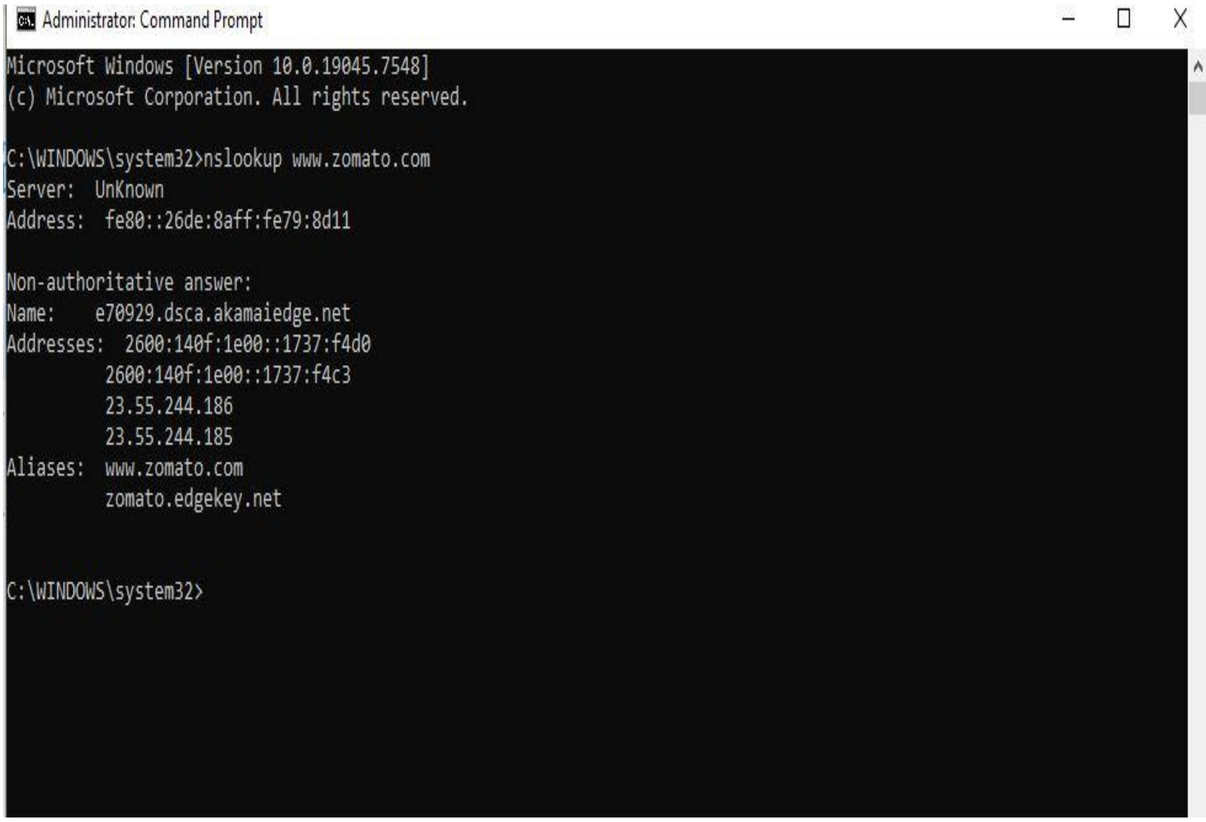
2. **Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.**



Intermediate hops shown in the tracert

- From this screenshot of intermediate hops shown in the output.
- Hop 1 : Local router – 192.168.1.254
- Hop 2 : ISP router – 10.230.192.1
- Hop 3 : ISP internal router – 10.111.24.5
- Hop 4 : ISP/Core network router – 136.232.112.109
- Hop 5 : Request timed out
- Hop 6 : Request timed out
- Hop 7 : Akamai edge server – 23.64.140.129
- Hop 8 : Destination server – 23.64.140.250

3. **Use the nslookup tool to find the IP address of www.zomato.com and record the result.**



4. **Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.**

```
Administrator: Command Prompt

C:\WINDOWS\system32>ping 192.168.1.9

Pinging 192.168.1.9 with 32 bytes of data:
Reply from 192.168.1.9: bytes=32 time=120ms TTL=64
Reply from 192.168.1.9: bytes=32 time=202ms TTL=64
Reply from 192.168.1.9: bytes=32 time=217ms TTL=64
Reply from 192.168.1.9: bytes=32 time=120ms TTL=64

Ping statistics for 192.168.1.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 120ms, Maximum = 217ms, Average = 164ms

C:\WINDOWS\system32>
```

```
Administrator: Command Prompt

C:\WINDOWS\system32>ping 192.0.0.4

Pinging 192.0.0.4 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.0.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\WINDOWS\system32>
```

- I tested connectivity to another device on my local network using the ping command. When the device was online, it replied successfully with low response time, confirming that it was reachable. When the device was offline or disconnected from the network, the ping command returned request timed out, indicating that no response was received and the device could not be reached over the network.

5. **Create a short report (5-7 lines) summarizing which tool (ping, traceroute, and nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.**

Short Report about Network diagnostic Tools

- Website not loading: Ping is the most useful tool to check if the destination server is reachable. In my test, when the target device was offline, the ping command returned "Request timed out" indicating the device could not be reached.
- Slow connection: traceroute is the best tool because it shows every network hop and helps identify where delay occur. In my test to www.flipkart.com multiple intermediate hops were displayed, allowing the network path to be analyzed.
- Suspecting DNS issues: Nslookup is the most useful tool because it verifies whether a domain name is being translated into an IP address correctly. For example, running nslookup www.zomato.com should return the website's IP address; if it fails, the issue is likely with DNS resolution.
- Overall, ping checks connectivity, traceroute identifies network path and delays and nslookup confirms whether DNS name resolution is working correctly.